



RNKE-Q

A proof-carrying verification layer for quantum computation

Every quantum job and every error-correction round gets a fail-closed, cryptographically chained accept / correct / defer / reject verdict — grounded in machine-checked theorems, already running on IBM hardware across three backends.

The problem: a quantum answer is not checkable



Noisy hardware returns numbers. Today, 'result' means 'whatever the backend returned.'

1 **No gate between QPU and user**
A job comes back; nothing independently decides whether it should be accepted, corrected or rejected.

2 **QEC results are 'trust me'**
Error-correction decoders output a correction; there is no re-audited, tamper-evident proof that the correction was lawful.

3 **No audit trail**
Vendors, labs and regulators cannot later verify which backend, calibration, shots and thresholds produced a claim.

4 **Claims drift**
Results get promoted from 'observed on one backend' to 'validated' with no boundary — a credibility problem for the whole field.

What RNKE-Q is



Recognition-Null Kernel Engine: nothing becomes a result merely because it arrived.

1 • Typed verdicts

Every QPU job, QEC round or sensor read ends in exactly one of:

COMMITTED — residue zero, accepted

CORRECTED — bounded residue, correction applied and re-audited to zero

DEFERRED — evidence disagrees; held, re-extraction requested

CUT — outside the declared model; rejected

State advances only on COMMITTED / CORRECTED. Fail-closed by construction.

2 • Proof-carrying certificates

Each verdict is a certificate: pinned circuit hash, backend, calibration epoch, shots, declared model and thresholds, exact-arithmetic residue, hash-chained receipt to its parent, and an explicit claim boundary.

Mandatory controls: a planted failure the verdict must reject, and a renaming control — the same computation on random data — so identities are never reported as findings.

3 • Theorem-grounded

Verdicts execute finite theorems of the Recognition framework: seam-integer threshold count, square-sourced (SOS) residue, sign localisation — foundational cores machine-checked in Lean 4.

Same kernel already ships in two other products: SETU (secure carrier-blind messaging) and the RNKE Agent Firewall (AI-agent action gate). One grammar, three domains.

Evidence L1 & L2: replication + exact QEC gate



All certificates and pins are in the repository; reviewers can re-run every check without credentials.

L1 • Operator-memory benchmark on 3 IBM backends

7-qubit circuit: 3 visible qubits, 3 memory-ledger qubits, 1 readout.
Anticommuting operator order stored in ledgers; selective erase, full erase, projection-erased, commuting and order-erased controls; reduced-model comparison.

Runs 14 Jul 2026: ibm_kingston · ibm_fez · ibm_marrakesh
All 6 predeclared per-run gates passed on every backend
Classification: CROSS_BACKEND_REPLICATION
Signed certificates, SHA-256 manifests, offline evaluator, CI self-tests.

Claim: reproducible quantum representation of decomposable operator-order memory. Not quantum advantage, not a device.

L2 • Exact QEC gate + theorem binding

9-qubit Shor code under the RNKE gate: 4 logical states $\times$ (clean + 27 single-Pauli + 3 two-error controls + disagreement) = 128 cycles — every clean state COMMITTED, every single error CORRECTED and re-audited, no two-error case ever COMMITTED, disagreement DEFERRED, receipt chain intact. Certificate pinned.

Theorem binding on measured IBM shots (ibm_marrakesh, 4 cases $\times$ 4096): seam integer k_Σ from stabilizer expectations equals injected error weight in all 4 cases (0, 1, 1, 4); stabilizer Gram certified PSD with explicit SOS weights; residue typed by exact inertia; renaming control shows which parts are identities.

Evidence L3: hardware syndromes + control loop



Where the kernel meets a real control loop.

17-qubit syndrome extraction on ibm_marrakesh

9 data + 8 ancilla qubits, 4 cases $\times$ 4096 shots (no error, X on q0, Z on q0, Y on q4).

Expected 8-bit syndrome is the unique modal outcome in 4 / 4 cases; per-stabilizer-bit accuracy $\approx$ 0.80–0.87.

Status: one non-repeated round — syndrome verified on hardware; recovery on hardware not yet claimed. Three-round repeated-extraction run submitted (job da2sogk3...).

Recognition-adaptive dynamical decoupling

Built: Ramsey / Hahn / CPMG-n / XY4-n arms plus an RNKE-selected schedule per qubit (from $\delta_q = \log 2T_1/T_2$, readout error, syndrome confidence) and a random-schedule renaming control. Predeclared fit and CI; headline only if adaptive beats best fixed schedule with CI separation. Status: submitted on ibm_marrakesh, 6 qubits (job da2std6...). Result $\rightarrow$ certificate, whichever way it falls.

How a reviewer checks this in 10 minutes

git clone $\rightarrow$ python -m unittest (45 tests, no credentials) $\rightarrow$ every EXPECTED_*.sha256 pin regenerates byte-for-byte $\rightarrow$ IBM job IDs are public and re-fetchable. The Qmet evidence bundle binds all components into one hash-chained index that composes provenance, not claim strength.

What we do not claim · what is under test



This slide is the product: a verdict is only as valuable as its boundary.

Not claimed

- Quantum advantage or speed-up
- Fault tolerance or an error threshold
- A validated physical device or sensor (the NV/ODMR component is a digital twin only)
- Any result on the Riemann Hypothesis or K_0 from this work
- 'Qmet-validated' or any institutional endorsement

Evidence classes are never silently upgraded: definition < derivation < computational verification < hardware observation < replication < physical validation. A bundle's class is the minimum of its members.

Hypotheses under test (predeclared)

H1 · Thermodynamic-response decoherence: an admissible (Lindblad) form of the recognition dissipator predicts that the stiffness residue $\delta_q = \log(2T_1/T_2)$ anti-correlates with per-stabilizer syndrome confidence on one backend epoch. Test runner written; fails closed (ABSTAIN) until the calibration snapshot is pinned. Verdict will be published either way.

H2 · Adaptive DD selection effect (slide 5).

Both are experiments, not results. They appear here so the reviewer sees the method, not a promise.

Intellectual property and founder



Three PCT applications; publication and replication discipline in the open.

Patents (PCT, pending, not yet published)

PCT/IB2026/058465 — Recognition-Aware Sensor, Logic and Control Apparatus (filed 10 Aug 2026). Recognition-before-action architecture: blindness evaluator, verification gates, source-bound proof-carrying validation, fail-closed controller — the IP anchor of RNKE-Q.

PCT/IB2026/051695 — Entropy-Governed Transistor, Logic Gate and Computation System with Invariant-Validated State Transitions (filed 21 Feb 2026).

PCT/IB2025/060887 — Unified Framework for Emergent Thermodynamic and Quantum-Gravitational Systems (filed 26 Oct 2025, priority 6 Aug 2025).

Public: arXiv:2603.20773 (cond-mat.stat-mech); Recognition-Kernel-Framework repo with Lean-verified theorems (Zenodo DOI 10.5281/zenodo.21760119); public red-team challenge for the kernel (4 attacks, none succeeded).

Founder

Parveen (Monty Dabas)
Founder & Director, Celextrix Pvt Ltd (inc. 4 Sep 2025, Delhi)
B.Tech 2012 · M.Tech 2016, Delhi Technological University

Built end-to-end: recognition kernel, SETU messenger (live, first real handshake 17 Aug 2026), Agent Firewall demo, quantum benchmark + QEC lab, certificates, Lean proofs.

Solo founder; engineering scales through the incubator. Seeking a quantum faculty mentor at IIT Bombay.

Business model, roadmap and ask



One product — Certified Quantum Execution — three buyers. Two-stage programme.

Business model

Per-job / per-round certification API for QPU users and vendors.
On-prem licence for defence, government and regulated labs.
Kernel licensing to QEC-decoder and control-stack vendors.
Same kernel sells today as the RNKE Agent Firewall (enterprise AI-agent security) — revenue while the quantum market forms.

Stage 1 • ₹50 L • months 0–12

- Replicate on a 2nd QPU vendor
- Multi-round QEC + recovery-on-hardware certified
- Qiskit / OpenQASM plugin
- Independent evaluator runs the locked suite
- H1/H2 verdicts published
- PCT national phase (IN, US, EP)

Stage 2 • ₹3–5 Cr • months 12–30

Milestone-gated. Team of 6. Partner QPU for logical-qubit-era QEC certification; sensor certification with an NQM lab (physical, not digital twin); regulator-grade audit pilot; secure certificate transport via SETU.

The ask to SINE / IIT Bombay

Incubation (virtual acceptable) • Stage-1 seed ₹50 L or introduction to iDEX-SINE / NIDHI tracks • quantum faculty mentor (Qmet / NQM) • QPU and lab access • partner-incubator support for our live iDEX Open Challenge application.
We are raising ₹50 L now toward a ₹4 Cr programme. Everything on these slides is reproducible from the repository today.